

Homework 5 Written Report: Applied Adversarial Audit of COMPAS

Based on the executed notebook submission

This report summarizes four adversarial audit results on the ProPublica COMPAS dataset using the lecture pipeline: (1) PGD evasion, (2) fairness-targeted label-flip poisoning, (3) membership inference, and (4) a final governance reflection. The central finding is that standard aggregate metrics are not enough for high-stakes deployment. In this notebook, attacks that changed subgroup harm materially could leave AUC nearly unchanged, which means an auditor must evaluate both system security and distributional impact rather than accuracy alone (Vassilev et al., 2024).

1. PGD evasion audit

For logistic regression, increasing the perturbation budget epsilon caused false positive rates to rise sharply for both groups. At baseline, FPR was 0.281 for African-American defendants and 0.143 for Caucasian defendants (AIR = 1.961). Under PGD, those values rose to 0.569 and 0.370 at epsilon = 0.25, 0.791 and 0.560 at epsilon = 0.50, and 1.000 and 1.000 at epsilon = 2.0. Gradient-boosted trees were slightly less vulnerable at low-to-moderate budgets under the same adversarial examples: their FPRs were 0.553 and 0.351 at epsilon = 0.25 and 0.733 and 0.514 at epsilon = 0.50, compared with 0.569 and 0.370 and 0.791 and 0.560 for logistic regression. However, that difference disappears as attack strength grows. By epsilon = 2.0, both models collapse to FPR = 1.0 for both groups. Importantly, AIR never crosses 0.80 for either model; instead it moves toward 1.0 because the attack equalizes harm by making nearly everyone look high risk. So the models are not equally vulnerable in a narrow technical sense—GBT is somewhat more resistant early—but they are equally unsuitable for high-stakes deployment without stronger robustness controls, because both can be driven into catastrophic false positives. Because the notebook applies LR-crafted perturbations to both models, the GBT result is best interpreted as a transfer-style evasion evaluation, which is still realistic under the NIST black-box/gray-box threat model (Madry et al., 2018; Vassilev et al., 2024).

2. Poisoning loop with fairness monitoring

The poisoning results show the most serious governance problem in the notebook: fairness can worsen substantially while aggregate performance appears stable. When high-risk labels were flipped for African-American training records, test AUC stayed between 0.730 and 0.735 across poison rates from 0% to 30%, so the maximum AUC change was only -0.004. But AIR moved from a baseline of 1.961 to 3.010 at a 30% poison rate, meaning subgroup disparity became much worse even though standard validation would look normal. The extension that targets Caucasian defendants shows the same basic pattern in the plotted curves: AUC remains near baseline while fairness shifts by target group. This confirms the lecture claim that poisoning is hard to detect with ordinary accuracy monitoring. There is no distinct 'stealth zone' in the strict assignment sense, because AIR is already outside the acceptable [0.80, 1.25] band at baseline. Still, the attack is stealthy with respect to AUC, since model performance hardly moves while fairness does. A PSI-based drift monitor would also miss both attacks in this notebook, because the poisoning changes labels rather than features; the feature-level PSI check is effectively zero, so the input distribution looks stable even when subgroup harm changes. This is why NIST-style data provenance and training-pipeline integrity controls matter more here than dashboard accuracy checks (Vassilev et al., 2024).

3. Membership inference depth

The membership-inference results are much weaker than the poisoning results. The logistic-regression model had MI AUC = 0.5047 and the gradient-boosted tree had MI AUC = 0.4939, both essentially random. At the same time, the generalization gap differs sharply across models: -0.0077 for logistic regression and 0.0802 for the gradient-boosted tree. So in this experiment, a larger generalization gap did not translate into a stronger membership-inference attack, even though the theory predicts that overfit models often create a larger confidence gap between members and non-members (Shokri et al., 2017). Varying L2 regularization on the logistic-regression model also produced only a tiny privacy-utility tradeoff. Across $C = \{0.01, 0.1, 1.0, 10.0\}$, MI AUC stayed in a very narrow band from 0.5036 to 0.5069, while test AUC changed only from 0.7321 to 0.7346. The practical implication is that, in this notebook, regularization does not materially change privacy risk because the attack is already near chance. For this model family, the more urgent problem is not training-data leakage but fairness-sensitive manipulation of the pipeline or deployment inputs.

4. Reflection and recommendation

My single highest-risk finding is the label-flip poisoning attack. It is the most dangerous because it changes downstream fairness outcomes while evading the performance metrics that many organizations watch first. The notebook quantifies this clearly: the worst poisoning run moved AIR by +1.049 relative to baseline while changing AUC by only -0.004. A strong proactive mitigation is to harden the training pipeline with immutable data snapshots, restricted write access, label-audit checks, and provenance logging. A reactive mitigation is to monitor subgroup FPR and AIR in production and force escalation when those metrics move materially even if AUC remains flat. These controls do introduce tradeoffs. Stronger integrity controls slow retraining and increase operational overhead, and fairness-trigger monitoring can produce more reviews or temporary freezes. But in a criminal-justice system, that tradeoff is justified: silent integrity failures and hidden subgroup harm are more costly than slower model updates. The broader lesson is that a model can look accurate and still be unfit for deployment if it is brittle under attack or if its errors are redistributed in ways that standard dashboards do not reveal.

References

- Madry, A., et al. (2018). Towards deep learning models resistant to adversarial attacks. ICLR 2018.
- Shokri, R., et al. (2017). Membership inference attacks against machine learning models. IEEE S&P 2017.
- Vassilev, A., Oprea, A., Fordyce, A., & Anderson, H. (2024). Adversarial Machine Learning: A Taxonomy and Terminology of Attacks and Mitigations. NIST AI 100-2e2023.